# Machine Learning Laboratory

## Experiment No. 01

## Introduction to Machine Learning Workflow and Dataset Understanding

# 1 Aim

To understand the complete Machine Learning workflow and explore a machine learning dataset by identifying features, target variables, performing basic data analysis, and preparing the dataset for machine learning model development.

# 2 Objectives

After successful completion of this experiment, students will be able to:

- Understand different stages involved in a Machine Learning workflow.
- Explore and analyze the structure of a machine learning dataset.
- Identify independent variables and dependent variables.
- Perform basic exploratory data analysis.
- Understand the importance of data preprocessing in Machine Learning.
- Apply Train-Test Split for preparing datasets.
- Prepare datasets for implementing machine learning algorithms.

# 3 Prerequisites

Students should be familiar with:

- Python Programming
- NumPy Library

- Pandas Library
- Matplotlib Library
- Basic Statistics Concepts

# 4 Software Requirements

- Google Colaboratory / Jupyter Notebook
- Python 3.x
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

# 5 Theory

## 5.1 Understanding Machine Learning Dataset

A Machine Learning dataset is a collection of observations represented in the form of rows and columns. Each row represents an individual data sample, while each column represents an attribute or feature associated with that sample.

Before applying any machine learning algorithm, it is important to understand the structure, characteristics, and quality of the available data.

## 5.2 Components of a Dataset

A typical Machine Learning dataset consists of:

- **Samples:** Individual observations or records present in the dataset.
- **Features:** Input variables used by the machine learning model to learn patterns.
- **Target Variable:** The output variable that the model attempts to predict.

## 5.3 Dataset Exploration

Dataset exploration is the process of analyzing and understanding the available data before model development.

The important dataset exploration steps include:

- Checking the number of samples and features.
- Understanding data types of attributes.
- Identifying missing values.
- Observing statistical properties of numerical features.
- Understanding distribution of target classes.

## 5.4 Feature and Target Separation

In supervised learning, the dataset is divided into:

- **Feature Variables (X):** Input attributes provided to the model.
- **Target Variable (y):** Output that the model learns to predict.

## 5.5 Train-Test Split

Before training a machine learning model, the available dataset is divided into training and testing subsets.

- **Training Data:** Used for learning patterns from data.
- **Testing Data:** Used for evaluating model performance on unseen data.

In this experiment, the dataset understanding and preparation steps are performed before applying machine learning algorithms in subsequent experiments.

## 5.6 Importance of Data Preparation

Data preparation is an important step before applying machine learning algorithms. Real-world datasets may contain missing values, inconsistent formats, duplicate records, or categorical attributes.

Proper data preparation improves the quality of input data and helps machine learning algorithms learn meaningful patterns effectively.

## 5.7 Introduction to Scikit-learn

Scikit-learn is an open-source Python library used for implementing Machine Learning algorithms and related tasks. It provides efficient tools for data preprocessing, model development, model selection, and evaluation.

In this experiment, Scikit-learn is used to load the Iris dataset and perform train-test splitting.

## 5.8 Iris Dataset

The Iris dataset is a well-known benchmark dataset used for understanding basic concepts of Machine Learning. It contains measurements of iris flowers belonging to three different species.

The dataset consists of four input features and one target variable representing the flower species.

Table 1: Iris Dataset Information

| **Parameter** | **Description** |
|---|---|
| Number of Samples | 150 |
| Number of Features | 4 |
| Number of Classes | 3 |
| Target Variable | Species Class |

## 5.9 Features of Iris Dataset

The input features available in the dataset are:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

## 5.10 Target Classes

The target variable contains three classes:

- Setosa

- Versicolor
- Virginica

The objective of this experiment is to explore the dataset, understand its structure, separate input features and target variables, and prepare the data for implementing machine learning algorithms in subsequent experiments.

# 6 Procedure

The following steps are performed in this experiment:

1. Import the required Python libraries for data handling and analysis.
2. Load the Iris dataset using the Scikit-learn library.
3. Convert the dataset into a Pandas DataFrame for analysis.
4. Display the initial records of the dataset.
5. Explore the dataset structure using suitable data analysis functions.
6. Identify the independent variables (features) and dependent variable (target).
7. Analyze the dataset dimensions and statistical information.
8. Check the dataset for missing values.
9. Perform Train-Test Split to divide the dataset into training and testing subsets.
10. Observe the size and structure of training and testing datasets.

## 6.1 Expected Outcome

After completion of this experiment, students will be able to understand the structure of a machine learning dataset and prepare it for applying machine learning algorithms.

# 7 Practical Implementation

## 7.1 Step 1: Import Required Libraries

The required Python libraries are imported for data manipulation and analysis.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

## 7.2 Step 2: Load Iris Dataset

The Iris dataset is loaded using the Scikit-learn library.

```
from sklearn.datasets import load_iris

iris = load_iris()
```

## 7.3 Step 3: Display Dataset Components

The different components available in the Iris dataset are displayed.

```
print("Feature Names:")
print(iris.feature_names)

print("\nTarget Classes:")
print(iris.target_names)
```

The output displays information such as feature names, target values, and dataset description.

## 7.4 Step 4: Create DataFrame

The feature data is converted into a Pandas DataFrame for further analysis.

```
df = pd.DataFrame(
    iris.data,
    columns = iris.feature_names
)
```

## 7.5 Step 5: Add Target Variable

The target variable representing flower species is added to the DataFrame.

```
df["Species"] = iris.target
```

## 7.6 Step 6: Display Dataset Records

The first few records of the dataset are displayed.

```
df.head()
```

## 7.7 Step 7: Dataset Exploration

Basic information and statistical summary of the dataset are obtained.

```
df.info()
```

```
df.describe()
```

```
df.shape
```

## 7.8 Step 8: Check Missing Values

The presence of missing values in the dataset is checked.

```
df.isnull().sum()
```

## 7.9 Step 9: Check Class Distribution

The number of samples belonging to each target class is observed.

```
df["Species"].value_counts()
```

## 7.10 Step 10: Separate Features and Target

The dataset is divided into independent variables (features) and dependent variable (target).

```
X = df.iloc[:, :-1]
```

```
y = df.iloc[:, -1]
```

### 7.11 Step 11: Train-Test Split

The dataset is divided into training and testing subsets using Scikit-learn.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

### 7.12 Step 12: Display Dataset Sizes

The dimensions of training and testing datasets are displayed.

```
print("Training Data :", X_train.shape)

print("Testing Data :", X_test.shape)
```

## 8 Dataset Visualization

Data visualization helps in understanding the distribution and relationship between different features of the dataset.

In this experiment, a scatter plot is used to visualize the relationship between sepal length and sepal width of Iris flowers.

```
plt.figure(figsize=(6,4))

plt.scatter(
    df["sepal length (cm)"],
    df["sepal width (cm)"]
)

plt.xlabel("Sepal Length (cm)")
plt.ylabel("Sepal Width (cm)")
plt.title("Iris Dataset Visualization")

plt.show()
```

# 9 Observation

The following observations are recorded after executing the experiment:

- Total number of samples present in the dataset.
- Number of input features available in the dataset.
- Number of target classes present.
- Structure and information of the dataset using `df.info()`.
- Statistical characteristics of features using `df.describe()`.
- Presence of missing values in the dataset.
- Size of training and testing datasets after Train-Test Split.

# 10 Result

The Iris dataset was successfully loaded, explored, and prepared for Machine Learning. The independent features and target variable were identified, and the dataset was divided into training and testing subsets using Train-Test Split.

The prepared dataset can be used for implementing Machine Learning algorithms in subsequent experiments.

# 11 Think and Observe

1. Why do we separate features and target variables before training a machine learning model?
2. Why is a dataset divided into training and testing subsets?
3. What happens if a model is trained and tested using the same dataset?
4. Why is data preprocessing important before applying machine learning algorithms?
5. What information is obtained from functions such as `df.info()` and `df.describe()`?